US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication 20250268268
Kind Code A1
Publication Date August 28, 2025
Inventor(s) TASSY, SR.; Tomas et al.

HIGH VOLUME SCORING SYSTEM FOR BREAD DOUGH

Abstract

A high volume scoring system for bread dough is provided with a conveyor table that includes a first scoring apparatus and a second scoring apparatus that is located downstream from the first scoring apparatus to permit longitudinal and angled scoring of dough pieces. Trays containing unbaked dough pieces are loaded onto the conveyor and shuttled downstream along the conveyor to selectively align the trays beneath the first or second scoring apparatus depending on whether diagonal or longitudinal score lines are desired on the dough pieces. The first scoring apparatus is configured to make a series of diagonal score lines on the dough pieces positioned beneath the first scoring apparatus using a first series of blades. The system can also convey the trays containing the unbaked dough pieces downstream along the conveyor to align the trays beneath the second scoring apparatus. The second scoring apparatus is configured to make longitudinal score lines on the dough pieces positioned beneath the second scoring apparatus using a second series of blades. Once the dough pieces have been scored, they are conveyed by the system into an oven for baking.

Inventors: **TASSY, SR.; Tomas (East Dundee, IL), SUTO; Antal (East Dundee, IL)**

Applicant: **T&T CONSULTING AND ENGINEERING, INC.** (East Dundee, IL)

Family ID: **1000008478307**

Appl. No.: **19/060205**

Filed: **February 21, 2025**

Related U.S. Application Data

us-provisional-application US 63556629 20240222

Publication Classification

Int. Cl.: **A21C11/12** (20060101); **A21C9/08** (20060101)

U.S. Cl.:

CPC **A21C11/12** (20130101); **A21C9/08** (20130101);

Background/Summary

PRIORITY CLAIM [0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application Ser. No. 63/556,629 filed Feb. 22, 2024, which is expressly incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE
[0002] The present disclosure relates generally to baking equipment, and more specifically to baking equipment used in the commercial baking industry.
BACKGROUND
[0003] The baking industry, specifically the commercial baking industry, produces numerous baked goods such as loaves of bread, hamburger buns, hot dog buns, pizza crust, and pita bread, among other items. Several steps need to be performed to bread dough during preparation. Bread scoring of proofed dough pieces is the final step prior to baking certain bread products. Scoring creates an incision on the skin/surface of the dough, and can have a direct impact on overall appearance of the bread loaf. In small scale production, this process is typically done manually with sharp metallic blades (lames in French), while water jets are typically used in large scale scoring operations. Manual scoring adds unnecessary time, cost, and complexity to the baked process and is too slow for large scale operations. Use of water jets requires a high pressure water source and is highly complex and costly to implement. There is a need for a high volume scoring system for bread dough that will allow for the desired scoring of dough pieces without slowing down the overall commercial baking process to reduce the overall cost of the scored baked goods.
SUMMARY
[0004] The present disclosure may comprise one or more of the following features and combinations thereof.
[0005] In illustrative embodiments, the present disclosure is directed to a high volume scoring system for bread dough. The scoring system a conveyor table that includes a first scoring apparatus and a second scoring apparatus that is located downstream from the first scoring apparatus. The present arrangement allows for both longitudinal and angled scoring of the dough pieces.
[0006] In illustrative embodiments, trays containing unbaked dough pieces are loaded onto the conveyor and shuttled downstream along the conveyor to align the trays beneath the first scoring apparatus. The first scoring apparatus is configured to make a series of diagonal score lines on the dough pieces positioned beneath the first scoring apparatus.
[0007] In illustrative embodiments, the conveyor can also convey the trays containing the unbaked dough pieces downstream along the conveyor to align the trays beneath the second scoring apparatus. The second scoring apparatus is configured to make longitudinal score lines on the dough pieces positioned beneath the second scoring apparatus. Once the dough pieces have been scored, they are conveyed into an oven for baking.
[0008] These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a perspective view of a high volume scoring system of the present disclosure

showing a linear conveyor for conveying trays of unbaked dough pieces, a first scoring apparatus and a second scoring apparatus positioned downstream from the first scoring apparatus;

[0010] FIGS. **2-7** illustrate the first scoring apparatus;

[0011] FIG. **2** is a perspective view of the first scoring apparatus fully assembled;

[0012] FIG. **3** is a side elevational view of the first scoring apparatus of FIG. **2** showing a series of serrated blades used to score dough pieces positioned beneath the blades;

[0013] FIG. **4** is a perspective view of the first scoring apparatus with a top plate removed showing the middle plate and the bottom plate and also showing the diagonal slide rails;

[0014] FIG. **5** is a perspective view of the first scoring apparatus with the top plate and middle plate removed and showing the bottom plate and showing the transverse slide rails that are transverse to the direction of movement of the trays on the conveyor;

[0015] FIG. **6** is a perspective view of the first scoring apparatus with the top, middle and bottom plates removed and showing the plurality of blades mounted in blade holders that are in three unit blocks;

[0016] FIG. **7** is a close up perspective view of FIG. **6** showing the blades passing through a blade cleaner plate that removes unwanted dough from the blades;

[0017] FIG. **8** is a perspective view of the second scoring apparatus showing the apparatus inside of a safety enclosure and secured to the linear conveyor;

[0018] FIG. **9** is a perspective view of the overhead shuttle of the second scoring apparatus and also showing the blade wiper assembly in front of the blades;

[0019] FIG. **10** is a side perspective view of the second scoring apparatus showing the scoring blade assembly secured to a linear actuator used to raise and lower the blades and also showing the blade wiper assembly;

[0020] FIG. **11** is another perspective view of the wiper assembly and the blade assembly;

[0021] FIG. **12** is another perspective view of the wiper assembly with one of the wiper blocks removed to show the position of the blade;

[0022] FIG. **13** is a top view of the second scoring apparatus showing a tray of dough pieces positioned below the second scoring apparatus and showing the dough pieces each including a longitudinal slit added to the dough pieces by the blades;

[0023] FIG. **14** is another perspective view of the second scoring apparatus showing one of the wiper blocks **108** removed from a wiper bar used to secure the wiper blocks;

[0024] FIG. **15** is a perspective view of the conveyor used to convey the trays of dough pieces beneath the first and second scoring apparatuses;

[0025] FIG. **16** is a side elevational view of the conveyor showing the linear cylinder used to covey a shuttle along a frame of the conveyor;

[0026] FIG. **17** is an end view of the conveyor;

[0027] FIG. **18** a second side elevation of the conveyor;

[0028] FIG. **19** is a top view of the conveyor;

[0029] FIG. **20** is a perspective view of the conveyor with portions removed to show the shuttle that is moved linearly along the frame by the linear cylinder;

[0030] FIG. **21** is a perspective view of a pusher component of the conveyor;

[0031] FIG. **22** is a perspective view of an alternative embodiment of the blade arrangement used in the first scoring apparatus showing the plurality of razor blades mounted in blade holders secured to a support bar;

[0032] FIG. **23** is a side elevational view of the second scoring apparatus showing the alternate embodiment of the blade assembly showing a blade holder coupled to a blade bar wherein the blade holder is holding a razor blade;

[0033] FIG. **24** is a perspective view of the blade assembly of FIG. **23** showing a razor blade secured into a blade holder and blade arm; and

[0034] FIG. **25** is a perspective view of the first scoring apparatus showing the alternative embodiment of the diagonal cut blade assembly having a series of blades secured to blade holders and used to cut diagonal score lines on the dough pieces.

DETAILED DESCRIPTION

[0035] For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

[0036] A high volume scoring system **10** for scoring bread dough pieces **14** is shown in FIG. **1**. The high volume scoring system **10** is part of a commercial baking process and is a portable unit that can be added to any baking line. The scoring system **10** is integrated into a commercial baking process that includes a tunnel oven (not shown) located downstream from the high volume scoring system **10**. Rows of baking trays **12** containing individual dough pieces **14** are placed on a linear conveyor **16**, which forms part of the scoring system **10**. Linear conveyor **16** is configured to convey the baking trays **12** to either a first scoring apparatus **18** or a second scoring apparatus **20**, which can be positioned either upstream or downstream from the first scoring apparatus **18**. The first scoring apparatus **18** is configured to place diagonal score lines **22** across the dough pieces **14**. The second scoring apparatus **20** is configured to place longitudinal score lines **24** along the length of the dough pieces **14**. First and second scoring apparatuses **18**, **20** can either be fixed in position relative to conveyor **16** or, for higher speed installations, can be configured to shuttle along the conveyor **16** with the movement of the trays **12** to score dough pieces **14** as they are moving along conveyor **16**.

[0037] First scoring apparatus **18** of high volume scoring system **10** is positioned downstream from the start of conveyor **16** and has the capability to score fifteen or more dough pieces **14** sequentially. While fifteen dough pieces are shown, the first scoring apparatus **18** can sequentially score any number of dough pieces **14** as desired. First scoring apparatus **18** includes a series of serrated blades **26** that extend downwardly at an angle to horizontal, as shown in FIGS. **2** & **3**. Blades **26** are configured to move along three axes of movement to properly score the dough pieces **14**.

[0038] First scoring apparatus **18** of high volume scoring system **10**, from top to bottom, includes a top plate **28**, a spaced apart middle plate **30**, a cross plate **31**, and a spaced apart lower plate **32**, as shown in FIGS. **2** and **3**. While top plate **28**, middle plate **30**, cross plate **31**, and lower plate **32** are described as plates, any other support structures or members can be used.

[0039] The top plate **28** of high volume scoring system **10** includes a linearly actuated first cylinder **34** that is secured to top plate **28** and includes a bracket **36** that extends through a slot **38** formed in the top plate **28**, as shown in FIGS. **2** and **3**. Top plate **28** is generally fixed in position. Bracket **36** is secured to middle plate **30** and is configured to move middle plate **30** along a first set of rail supports **42** by use of rails **40**, as shown in FIG. **4**. Actuation of first cylinder **34** causes movement of the middle plate **30**, cross plate **31**, bottom plate **32**, and the blades **26**, forty five (45) degrees to the direction of movement of trays along conveyor **16** to apply a diagonal score lines on dough pieces **14**. Rails **40** are slidably coupled to rail supports **42**. Rail supports **42** are secured to top plate **28** by use of fasteners. While the rails **40** are preferably oriented at 45 degrees to the direction of the conveyor **16**, it is contemplated that rails can be oriented at other desired angles from about 25 degrees to about 60 degrees, for example. Middle plate **30** is connected to lower rail supports **44** located proximate the bottom plate **32** by use of connectors **46**.

[0040] Cross plate **31** of high volume scoring system **10** is positioned below middle plate **30** and is used to move blades **26** in a linear direction that is perpendicular to the direction of the trays **12** along the conveyor **16**. Linear movement of the blades **26** is necessary when more than one diagonal score line is required on the dough pieces **14**. Once the first diagonal score line is cut, blades **26** are moved perpendicularly to the length of the conveyor so that an additional diagonal score line can be added to the dough pieces **14**. Cross plate **31** is moved by a second cylinder **48** in a linear direction that is perpendicular to the movement of the conveyor **16**. In other words, the second cylinder **48** moves the cross plate **31** transversely to the direction of the conveyor **16**. Second cylinder **48** is secured to bottom plate **32** by use of bracket **50**.

[0041] Cross plate **31** of high volume scoring system **10** is also coupled to a series of transversely mounted blade bars **52**, as shown in FIGS. **4** & **5**. Blade bars **52** are spaced apart from each other and move together with cross plate **31** when second cylinder **48** is activated. Cross plate **31** also is coupled to a pair of linear actuators **54**. Linear actuators **54** are coupled to each end of the cross plate **31** and are also coupled to the bottom plate **32**. Linear actuators **54** are designed to vertically raise and lower the cross plate **31** and blades **26** vertically from a first elevation to a second elevation, which is higher than the first elevation. In the first elevation, the blades **26** are not in contact with the dough pieces **14**. When in the second elevation, the blades **26** are in contact with the dough pieces **14** and the first cylinder **34** can be activated so that blades **26** can form diagonal score lines in dough pieces **14**.

[0042] Blades **26** are secured to blade bars **52** by use of blade holders **56**, as shown in FIG. **6**. There is one blade holder **56** for each blade **26** used to score the dough pieces **14**. Blade holders **56** include a lower leg **58** and an upper leg **60**. Lower leg **58** include a clamp member **62** that is used to secure blade **26** to lower leg **58** of blade holder **56**. This allows blades **26** to be replaced and/or adjusted as needed. The upper leg **60** of the blade holder **56** is configured to be secured to the blade bars **52** by use of fasteners. Lower leg **58** of blade holder **56** is shown as secured to upper leg **60** at an angle of about 45 degrees. While a fixed angle is shown, it is completed that lower leg **58** can be pivotally coupled to upper leg **60** so that any desired angle can be used, such a 25 degree or a 30 degree angle, for example.

[0043] In another embodiment, the blades **26** can be configured to each rotate about an axis of rotation. In this arrangement, the blade holders **56** would each include a bearing to allow the blades **26**. In this arrangement, it would be preferred to use a thicker plate with **2** bearings for each cutting blade holder support. The blade holders would then be interconnected by use of a linkage assembly. A common rod would be used to connect the linkage assembly to an air cylinder, which would allow an operator to rotate the blades to allow for straight or diagonal cuts.

[0044] Linear actuators **54** are coupled to bottom plate **32** and to cross plate **31**. Outer housing **58** is secured to cross plate **31** and inner member **60** is coupled to bottom plate **32**. Outer housing **58** is designed to extend away from inner member **60** to elevate cross plate **31**, blade bars **52** and blades **26**. Linear actuators **54** move cross plate **31**, blade bars **52**, blade holders **56**, and blades **26** in a vertical direction.

[0045] High volume scoring system **10** also include cleaner plate **63**, as shown in FIGS. **6** & **7**. Cleaner plate **63** is secured to bottom plate **32** by a series of posts **64**. Cleaner plate **63** also includes a series of slits **66** formed in the cleaner plate **62**. Slits **66** are sized to allow for the passage of blades **26**. After the blades **26** slit the dough pieces **14**, they are retracted upward so that cleaner plate **63** can remove any residual dough that may have clung to the blades **26** to, in effect, squeegee or wipe the blades of any dough residue. This arrangement limits the amount of maintenance that needs to be done to high volume scoring system **10**.

[0046] In use, a baking tray **12** that include a series of dough pieces **14** is positioned by conveyor **16** beneath the first scoring apparatus **18**. Once in position, linear actuators **54** lower blades **26** and first cylinder **34** moves blades **26** in a diagonal direction from a first position to a second position so that diagonal score lines are made on the dough pieces **14**. Next, blades **26** are raised by linear actuators **54** and first cylinder **34** moves blades back from the second position back to the first position. Next, second cylinder **48** moves blades **26** transversely to the conveyor **16**. Linear actuators **54** again lower the blades **26** and first cylinder **34** moves blades **26** in a diagonal direction from the first position to the second position so that second diagonal score lines are made in the dough pieces **14**. After the second score lines are cut, linear actuators **54** raise blades **26**, which are wiped clean as the blades **26** move through slits **66** in cleaning plate **62**.

[0047] Second scoring apparatus **20**, which is positioned downstream from first scoring apparatus

18 includes a housing **100** that is secure to conveyor **16**, as shown in FIG. **8**. Second scoring apparatus **20** is configures to place scoring lines onto a series of dough pieces **14** positioned on a tray **12** positioned beneath second scoring apparatus **20**. Second scoring apparatus **20** is configured to score all of the dough pieces **14** sequentially and then self-clean blades **102** that are used for the scoring operation. Second scoring apparatus **20** includes an overhead track **104** that is arranged perpendicular to the direction of the conveyor **16** and a shuttle **106** that is configured to be moved along the track **104** by a pneumatic cylinder (not shown).

[0048] Second scoring apparatus **20** includes a row of serrated cutting blades **102** positioned behind a row of blade wiper blocks **108**, as shown in FIG. **9**. Blades **102** are each secured to individual blade holders **110**. Blade holders **110** are each mounted onto a longitudinal blade bar **112**. Blade holders **110** are independently adjustable along blade bar **112** and are locked into position with set screws **114**. This arrangement allows the blade holders **110** and blades **102** to be independently adjustable with respect to one another and can be removed from the blade bar **112** altogether for servicing or to reduce the number of blades **102**. In the present disclosure, four blades are shown to score dough pieces **14** on a tray **12**. While four blades **102** are shown, three blades or five blades could also be used, for example.

[0049] Blade bar **112** is coupled to bracket **118**, which, in turn, is coupled to a vertically oriented linear actuator **120**, as shown in FIG. **9**. Linear actuator **120** is coupled to an arm **122**, which is secured to shuttle **106** to allow arm **122**, linear actuator **120**, bracket **118**, blade bar **112** and blades **102** to move perpendicular to movement of conveyor **16** by movement of shuttle **106** along overhead track **104**. Linear actuator **120** is configured to move bracket **118**, blade bar **112**, blade holders **110** and blades **102** vertically from a first position where the blades **102** are is position to score the dough pieces **14** and a second position where the blades **102** are elevated so they are no longer making contact with the dough pieces **14**.

[0050] Arm **122** at its lower end includes a leg **124** that is provided with an adjustable stop **126**. Adjustable stop **126** limits the downward movement of linear actuator **120** by making contact with bracket **118**. This arrangement allows for the adjustment of the lowermost extent of blades **102**. Blade holders **110** include blade retainers **128** that allow blades **102** to be removed and replaced as necessary. While blades **102** are secured by blade holder **110** at approximately 45 degrees, it is contemplated that other angles can be used and blade holder **110** can be made so that the angle can be adjusted by the user.

[0051] Second scoring apparatus **20** also includes a blade wiper assembly **130**, as shown in FIG. **10**. Blade wiper assembly **130** is configured to remove dough residual from blades **102** when retracted and moved upward away from dough pieces **14**. Blade wiper assembly **130** includes a pair of arms **132** that have an upper end coupled to shuttle **106** and a lower end coupled to a wiper bar **134**. Wiper bar **134** is horizontally oriented and is configured to accept adjustable brackets **136**. Brackets **136** are configured to secure wiper blocks **108**. Brackets **136** include adjustment thumb screws **138**. Screws **138** allow the wiper blocks **108** to be repositioned along wiper bar **134** to align wiper blocks **108** with blades **102** and then locked into position. Wiper blocks **108** each include a slit **140** that is sized to allow for the passage of blades **102** to wipe off any excess dough from the scoring process. Once blades **102** are elevated, shuttle moves blades **102** and wiper blocks **108** back to one side of the conveyor **16** to await the next tray of dough pieces **14**.

[0052] In use, linear actuator **120** starts with blades **102** in an elevated position and shuttle **106** moves blades **102** and blade wiper assembly **130** to one side of conveyor **16**. A tray **12** of dough pieces **14** is conveyed beneath second scoring apparatus **20**. Once the system has determined that the tray **12** is in the correct position, linear actuator **120** lowers the four blades **102**, as illustrated in the figures. Once the blades **102** are lowered, shuttle **106** moves along overhead track **104** and blades score dough pieces **14** sequentially along the tray until all of the dough pieces **14** are scored. Once the dough pieces **14** are scored, linear actuator **120** raises blade **102**, which are wiped off by wiper blocks **108** to remove any excess dough pieces **14**. While second scoring apparatus **20** is shown as fixed in the drawings, it can shuttle along with the trays **12** so that the operation can be continuous without requiring the trays to stop for scoring.

[0053] Conveyor **16** is an elongated structure used to convey trays **12** of dough pieces **14** under first and second scoring apparatuses **18**, **20**, as shown in FIGS. **15-21**. Conveyor **16** includes frame rails **142** supported by legs **144**. Leg sets are interconnected by longitudinal bar **146**. Conveyor **16** also include a shuttle **148** that is slidably connected to frame rails **142** by a set of rollers **150**. Rollers **150** are positioned within channels **152** formed in frame rails **142**.

[0054] Shuttle **148** of conveyor **16** includes a set of pushers **154**. Pushers **154** are used to push trays **12** along conveyor **16**. Conveyor also include a linear actuator in the form of a pneumatic cylinder **156**. Pneumatic cylinder **156** is coupled to longitudinal bar **146** and includes a bracket **158** that is connected to shuttle **148**. Cylinder **156** is configured to move shuttle **148** along frame rails **142** when cylinder **156** is extended. Shuttle **148** uses pushers **154** to push trays **12** the entire length of frame rails **142** of conveyor **16** into an oven. Pushers **154** include a pawl **160** that is pivotally coupled to bracket **162**. Pawl **160** is designed to engage trays **12** in a first down stream direction and pivot and go under upstream trays when moving in a second upstream direction.

[0055] In use, shuttle **148** of conveyor **16** is moved from an upstream position toward a downstream position by cylinder **156**. Pawls **160** of pushers **154** engage and push trays along frame rails **142** to the first and second scoring apparatuses **18**, **20** and then to an oven. Once the shuttle **148** reaches the end of its travel, cylinder **156** pulls shuttle **148** back upstream to engage new trays **12**. As shuttle **148** moves upstream, pawls **160** pivot about brackets **162** to allow pawls **160** to pass beneath trays **12**. Once past trays **12**, pawls **160** return to their original operating position so that they can again push trays **12** downstream.

[0056] FIGS. **22** and **25** are perspective views of an alternative embodiment of the scoring blade arrangement **200** used with the first scoring apparatus **18**. The scoring blade arrangement **200** uses razor blades **202** that are removably mounted in blade holders **204**. Blades **202** are secured to blade bars **206** by use of the blade holders **204**. There is one blade holder **204** for each blade **202** used to score the dough pieces **14**. Blade holders **204** include a lower leg **208** and an upper leg **210**. Lower leg **208** include a clamp member **212** that is used to secure blade **202** to lower leg **208** of blade holder **204**. This allows blades **202** to be replaced and/or adjusted as needed. Clamp member **212** includes a pivot **214** that allows the angle of the blade **202** to be adjusted for alignment purposes so that the proper depth score line is formed. The upper leg **210** of the blade holder **204** is configured to be secured to the blade bars **52** by use of fasteners. Lower leg **208** of blade holder **204** is shown as secured to upper leg **210** at an angle of about 45 degrees. While a fixed angle is shown, it is completed that lower leg **208** can be pivotally coupled to upper leg **210** so that any desired angle can be used, such a 25 degree or a 30 degree angle, for example.

[0057] FIGS. **23** and **24** show an alternate embodiment of the scoring blade arrangement **214** for the second scoring apparatus **20**. Second scoring apparatus **20** includes a row of replaceable razor scoring blades **216**. Scoring blades **216** are each secured to individual blade holders **218**. Blade holders **218** are each mounted onto a longitudinal blade bar **112**. Blade holders **218** are independently adjustable along blade bar **112** and are locked into position with set screws **220**. This arrangement allows the blade holders **218** and scoring blades **216** to be independently adjustable with respect to one another and can be removed from the blade bar **112** altogether for servicing or to reduce the number of scoring blades **216**. In the present disclosure, four blades are shown to score dough pieces **14** on a tray **12**. While four scoring blades **216** are shown, three blades or five blades could also be used, for example.

[0058] Blade holders **218** include blade retainers **222** that allow scoring blades **216** to be removed and replaced as necessary. While scoring blades **216** are secured by blade holder **218** at approximately 45 degrees, it is contemplated that other angles can be used and blade holder **218** can be made so that the angle can be adjusted by the user. Blade holders **218** include a pivot **224** that allows the angle of the scoring blades **216** to be changed with respect to the blade holders **218** for fine tuning.

[0059] The scoring device **10** is configured for scoring a series of dough pieces **14** arranged on a support surface, such as a tray **12**, with at least one score line **22**. The scoring device **10** includes a conveyor **16** is for conveying the dough pieces **14** in a longitudinal direction to a first scoring apparatus **18**. The first scoring apparatus **18** is configured for scoring the dough pieces **14** on the baking trays **12**. The first scoring apparatus **18** having a series of scoring blades **26**. The first scoring apparatus **18** includes a first actuator **34** configured for moving the scoring blades **26** in a first direction from a first position to a second position, which is generally a vertical movement toward and away from the dough pieces **14**.

[0060] The first scoring apparatus **18** includes a second actuator **48** configured for moving the scoring blades **26** in a second direction that is different from the first direction, which is generally diagonal to the direction of the conveyor **16**. Movement of the scoring blades **26** in the second direction causes the scoring blades **26** to form at least one score line **22** on each of the dough pieces **14**. The conveyor **16** is configured to convey the dough pieces **14** to the first scoring apparatus **18** for scoring and away from the first scoring apparatus **18** after the scoring of the dough pieces **14** is completed. The first scoring apparatus **18** also includes a blade wiper which can be in the form of a cleaner plate **66** that is configured to assist in wiping or cleaning dough from the blades after the scoring blades **26** score the dough pieces **14**.

[0061] The blade wiper is in the form of a cleaner plate **66** formed to include a series of slits **66**, wherein the blades are configured to move within the slits **66**. The scoring device **10** uses the conveyor **16** to convey and align the dough pieces **14** with the scoring blades **26**. The second actuator **48** is configured to move the scoring blades **26** in a direction that is diagonal to the longitudinal direction of the conveyor **16**. Alternatively, the scoring device **10** can be arranged so that the second actuator **48** is configured to move the scoring blades **26** in a direction that is perpendicular to the longitudinal direction of the conveyor **16**. The scoring blades **26** are generally arranged in a row and spaced apart from each other and are configured to move in unison when moved by the second actuator **48** to score the dough pieces **14**.

[0062] The scoring blades **26** can be used to apply multiple score lines on the dough pieces **14** so that each dough piece include more than one score line. The scoring device **10** also includes a second scoring apparatus **20** positioned downstream from the first scoring apparatus **18**. The second scoring apparatus **20** is configured for scoring the dough pieces **14**. For example, the first scoring apparatus **18** can apply diagonal score lines while the second scoring apparatus **20** can apply a score line that is perpendicular to the direction of travel of the conveyor **16**, for example. The second scoring apparatus **20** having a second series of scoring blades **102**. the second scoring apparatus **20** includes a first actuator **120** configured for moving the second series of scoring blades **102** in a first direction from a first position to a second position. The second scoring apparatus **20** includes a second actuator or shuttle **106** is configured for moving the second series of scoring blades **102** in a second direction that is different from the first direction. Movement of the second series of scoring blades **102** in the second direction causes the second series of scoring blades **102** to form at least one score line on each of the dough pieces **14**. The conveyor **16** is configured to convey the dough pieces **14** to the second scoring apparatus **20** for scoring and away from the second scoring apparatus **20** after the scoring is completed. The scoring device **10** can be programmed to selectively use the first or the second scoring apparatus to add score lines to the dough pieces **14**.

[0063] While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A device for scoring a series of dough pieces, arranged on a support surface, with at least one score line, the device comprising: a conveyor for conveying the dough pieces in a longitudinal direction; a first scoring apparatus configured for scoring the dough pieces, the first scoring apparatus having a series of scoring blades; a first actuator configured for moving the scoring blades in a first direction from a first position to a second position; a second actuator configured for moving the scoring blades in a second direction that is different from the first direction, wherein movement of the scoring blades in the second direction causes the scoring blades to form at least one score line on each of the dough pieces; wherein the conveyor is configured to convey the dough pieces to the first scoring apparatus for scoring and away from the first scoring apparatus after the scoring is completed.

2. The device of claim 1, further including a blade wiper configured to assist in wiping the blades after the scoring blades score the dough pieces.

3. The device of claim 2, wherein the blade wiper is in the form of a plate formed to include a series of slits, wherein the blades are configured to move within the slits.

4. The device of claim 1, wherein the device uses the conveyor to convey and align the dough pieces with the scoring blades.

5. The device of claim 1, wherein the second actuator is configured to move the scoring blades in a direction that is diagonal to the longitudinal direction of the conveyor.

6. The device of claim 1, wherein the second actuator is configured to move the scoring blades in a direction that is perpendicular to the longitudinal direction of the conveyor.

7. The device of claim 1, wherein the scoring blades are generally arranged in a row and spaced apart from each other and are configured to move in unison when moved by the second actuator to score the dough pieces.

8. The device of claim 1, wherein the scoring blades are used to apply multiple score lines on the dough pieces.

9. The device of claim 1, further including a second scoring apparatus positioned downstream from the first scoring apparatus, the second scoring apparatus configured for scoring the dough pieces, the second scoring apparatus having a second series of scoring blades; the second scoring apparatus includes a first actuator configured for moving the second series of scoring blades in a first direction from a first position to a second position; the second scoring apparatus includes a second actuator configured for moving the second series of scoring blades in a second direction that is different from the first direction, wherein movement of the second series of scoring blades in the second direction causes the second series of scoring blades to form at least one score line on each of the dough pieces; wherein the conveyor is configured to convey the dough pieces to the second scoring apparatus for scoring and away from the second scoring apparatus after the scoring is completed.

10. The device of claim 9, where in the device can selectively use the first or the second scoring apparatus to add score lines to the dough pieces.

11. A device for scoring a series of dough pieces, arranged on a support surface, with at least one score line, the device comprising: a first scoring apparatus configured for scoring the dough pieces, the first scoring apparatus having a series of scoring blades; a first actuator configured for moving the scoring blades in a first direction from a first position to a second position; a second actuator configured for moving the scoring blades in a second direction that is different from the first direction, wherein movement of the scoring blades in the second direction causes the scoring blades to form at least one score line on each of the dough pieces.

12. The device of claim 11, further including a blade wiper configured to assist in wiping the blades after the scoring blades score the dough pieces.

13. The device of claim 12, wherein the blade wiper is in the form of a plate formed to include a series of slits, wherein the blades are configured to move within the slits.
14. The device of claim 11, wherein the dough pieces are aligned with the scoring blades before being scored.
15. The device of claim 11, wherein the second actuator is configured to move the scoring blades in a diagonal direction.
16. The device of claim 11, wherein the second actuator is configured to move the scoring blades in a perpendicular direction.
17. The device of claim 11, wherein the scoring blades are generally arranged in a row and spaced apart from each other and are configured to move in unison when moved by the second actuator to score the dough pieces.
18. The device of claim 11, wherein the scoring blades are used to apply multiple score lines on the dough pieces.
19. A device for scoring a series of dough pieces, arranged on a support surface, with at least one score line, the device comprising: a conveyor for conveying the dough pieces in a longitudinal direction; a first scoring apparatus configured for scoring the dough pieces, the first scoring apparatus having a first series of scoring blades; a first actuator configured for moving the first series of scoring blades in a first direction from a first position to a second position; a second actuator configured for moving the first series of scoring blades in a second direction that is different from the first direction, wherein movement of the first series of scoring blades in the second direction causes the first series of scoring blades to form at least one score line on each of the dough pieces; wherein the conveyor is configured to convey the dough pieces to the first scoring apparatus for scoring and away from the first scoring apparatus after the scoring is completed; a second scoring apparatus positioned downstream from the first scoring apparatus, the second scoring apparatus configured for scoring the dough pieces, the second scoring apparatus having a second series of scoring blades; the second scoring apparatus includes a first actuator configured for moving the second series of scoring blades in a first direction from a first position to a second position; the second scoring apparatus includes a second actuator configured for moving the second series of scoring blades in a second direction that is different from the first direction, wherein movement of the second series of scoring blades in the second direction causes the second series of scoring blades to form at least one score line on each of the dough pieces; wherein the conveyor is configured to convey the dough pieces to the second scoring apparatus for scoring and away from the second scoring apparatus after the scoring is completed.
20. The device of claim 19, further including a blade wipers configured to assist in wiping the first and second series of scoring blades after the first and second series scoring blades score the dough pieces.